



A comprehensive review of geothermal energy extraction and utilization in oilfields

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ABSTRACT

By the nature of reliability, sustainability, abundant resource and minor impact on environment, geothermal energy is considered as one of the future resources to meet the world's growing energy demand. Harnessing geothermal energy from oil wells by existing assets, data and technologies features significant advantages over traditional geothermal wells, especially in reduced capital expenditure and operational risks. To raise awareness and promote discussions on oilfield geothermal development, this paper comprehensively characterized the oilfield associated geothermal resource, reviewed the current geothermal development in worldwide oilfields, pinpointed the existing challenges and introduced multidisciplinary technologies as possible solutions to unlock the potential and promote extensive development of geothermal energy from oilfields. Moreover, an integrated risk/opportunity assessment and management framework is proposed to improve the outcomes of diverse oilfield geothermal projects in practice.

1. Introduction

Geothermal energy is a renewable and sustainable energy and features weather independence, stable, operationally reliable, and environmentally friendly nature. It has been extensively studied to mitigate global warming, reduce air pollution and meet the need of global energy consumption. For decades, the utilizations of geothermal energy mainly in regions with high geothermal gradient with intense volcanic or hydrothermal activities, such as the Geysers in USA and Iceland in Europe. In addition to those areas, geothermal stored in hydrocarbon reservoirs also presents a great potential, not only because massive geothermal energy is existing in oil and gas reservoirs but also oilfields have enormous advantages to develop those geothermal energy.

Facing with declining reserves, increasing operation cost, volatile oil prices and green energy trend, oil and gas companies started to explore and utilize oilfield geothermal energy from existing assets, seeking for solutions to reduce operation cost, extend economic life of aging fields and achieve environmental and social benefits (Wang et al., 2016). As the reserve of geothermal resource in oil and gas reservoirs has been continuously studied, the considerable amount of geothermal storage are reported in worldwide oilfields (Bennett et al., 2012; Erdiuc et al., 2007; Li and Sun, 2014; Liu et al., 2015a; Wang et al., 2016). More importantly, oilfields possess unique economic and technical

advantages to utilize associated geothermal resource. The existing wellbore, surface facilities and useful data empower the oilfield geothermal project with reduced cost, minimized risk and significant convenience (Liu et al., 2015b, 2018; Wang et al., 2017). In return, these successful applications of oilfield geothermal projects also benefit the oilfield offset the overall operation cost, reduce the fossil fuel consumption and extend the economic life. Field operations have been conducted and made notable progress in oilfield applications of space heating, crude oil gathering and transportation, and geothermal power generation (Lund and Boyd, 2016; Wang et al., 2016; Xin et al., 2012).

Despite of some successful oilfield geothermal applications, there still exist challenges for the large-scale development of geothermal resource in oilfields. The challenges are recognized as low energy conversion efficiency, insufficient involvement, inadequate assessment and planning, unsound policies, and some other problems (Wang et al., 2016; Zarrouk and Moon, 2014). Both theoretical research and experimental tests are broadly carried out to seek for possible solutions to enhance the oilfield geothermal extraction and utilization. Thermo-electric technology is discussed as alternative geothermal extraction method to recover wider temperature range of geothermal resource (Liu et al., 2014; Wang et al., 2017). Improved geothermal extractions are reported by employing horizontal well (Cui et al., 2017; Feng et al., 2015; Zhang et al., 2014) and nanotechnology (Sui et al., 2017) into

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oilfield geothermal applications.

This paper aims to provide a comprehensive review on the features of oilfield geothermal resource, methods of geothermal extractions, techniques of geothermal utilization, current developments and challenges of such oilfield geothermal development. Outline are advanced technologies which are considered to enhance, spread and accelerate the geothermal development in oilfields. Moreover, an integrated risk/opportunity assessment and management framework is proposed to guide oilfield geothermal projects in practice. By delivering an integrated understanding of oilfield geothermal resource from multi-disciplinary perspectives, this paper aims to evoke new attention to oilfield geothermal resource, call for new idea to promote oilfield geothermal research, and provide the operators valuable information and practical methods to conduct oilfield geothermal project.

2. Characteristics of oilfield geothermal resource

2.1. Overview of geothermal resource in oilfields

In general, geothermal resources could be classified into three categories as low, intermediate and high temperature resources. This classification reflects the availability of different resources to be utilized under certain temperature, including spacing heating/cooling, industrial drying, power generation and other applications. Among a number of classification methods, here we will use the following categories (Chiasson, 2016).

- High temperature resource: above 150 °C.
- Intermediate temperature resource: between 90 °C and 150 °C.
- Low temperature resource: between 30 °C and 90 °C.

As a type of resources that co-exist with hydrocarbon in sedimentary basins, oilfield geothermal resource falls into the intermediate to low temperature category given that the produced fluids temperatures range between 65 °C and 150 °C (Liu et al., 2018). Numerous studies have shown the abundant geothermal resource in oil and gas reservoirs around the world. Augustine and Falkenstern (2012) mapped the temperature distribution of US lower 48 states at the depth of 3500 m (Fig. 1), which demonstrates the intermediate to high temperature in the areas with active oil and gas activities, such as Texas, Oklahoma, Louisiana, and North Dakota. Confirmed by the study of Tester et al. (2006), some wells in Texas, Oklahoma and Louisiana have relatively high temperatures (150 °C–200 °C) at the bottomhole depth. Only in the State of Texas, there are tens of thousands of wells with bottom hole temperatures over 121 °C, some up to 204 °C (Erdlac et al., 2007). Major

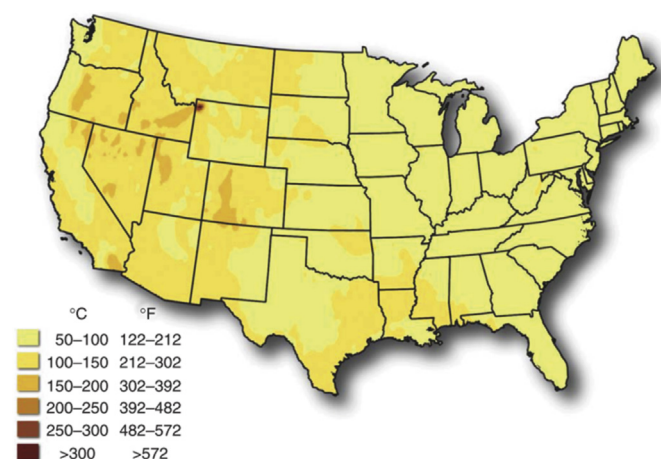


Fig. 1. Temperature map of US lower 48 states at depth of 3500 m (Augustine and Falkenstern, 2012).

Table 1

Geothermal resource storage in three oilfields in China (After Wang et al., 2016).

Oilfield	Total Geothermal Energy, EJ	Recoverable Geothermal Energy, EJ
Huabei	7099	306
Daqing	2905	89
Liaohe	1008	29
Total	11,012	424

petroliferous basins in China were also reported rich in geothermal resource, such as Daqing Oilfield, Liaohe Oilfield, and Huabei Oilfield, where the total reserves were up to 424 EJ (1 EJ = 10^{18} J) of recoverable geothermal resource as shown in Table 1 (Wang et al., 2016).

In view of the above summarized widespread distribution of massive oilfield geothermal resource, it is necessary and beneficial to understand features of geothermal resource along with oil and gas production, and thus provide better design to implement thermal recovery and utilizations.

2.2. Features of developing geothermal resource in oilfields

Compared to traditional geothermal fields, oilfields could provide tremendous advantages to develop geothermal energy through existing wellbores and facilities in a low cost and low risk manner. Details are elaborated as follows.

2.2.1. Cost-effectiveness

Producing geothermal energy from oil and gas wells is an economically efficient way. Zero or little drilling activity is needed since the wellbore with tested integrity is already there. On the other hand, the drilling cost of a traditional geothermal well could run as high as 50% of total cost (Barbier, 2002). The existing wellbore would eliminate the considerable cost and risk in drilling and completion operations. Moreover, existing surface infrastructures in oilfield, such as installed wellsite facilities, pipes, and service roads to wellsite, could further lower the initial investment. Even for those wells or facilities that may need little retrofit, only small amount investment is required to start up the project.

2.2.2. Minimized risk

From the long-term oil/gas exploration and production, sufficient data have been collected and analyzed. Besides the drilling and completion risk is reduced, the risk of reservoir uncertainties could be further minimized by capitalizing on these existing data for comprehensive reservoir characterization. For example, volumetric resource estimation method is widely applied to estimate thermal resource in geothermal reservoirs (Li and Sun, 2014). Fluid saturation is the crucial parameter for accurate estimation and is always assumed to be constant based on limited known data, which is obviously not true in oil and gas reservoir, where water saturation may increase, while the oil and gas saturations decrease to some extent. The alterations of fluid saturations during hydrocarbon production may induce errors in geothermal resource estimation in oil and gas reservoirs.

By courtesy of the oilfield database, which could not only provide the oil/gas/water saturations, but also the dynamic changes along with the production history, the operators are able to accurately evaluate the geothermal reserves at any interested point of production history, and help the decision making on the geothermal project.

2.2.3. Sufficient candidate wells

Oilfields have sufficient candidate wells for geothermal utilization, especially in mature oilfields, which have large number of high water cut wells as a result of water flooding and abandoned wells as a result of insufficient production. Those wells are losing or already lost economic

value, but they could be good candidates to make use of geothermal resource. Database of Texas Railroad Commission indicated that 10,370 wells are abandoned and plugged in 2016 in Texas. In China, there are 164,076 oil and gas wells, among which 76,881 wells have been abandoned (Li et al., 2015). Among sufficient candidate wells, operators could pick those high water-cut wells with high bottomhole temperature and large production rate, select those abandoned wells with high bottomhole temperature and reliable wellbore integrity, and retrofit for geothermal production, which could produce clean energy, awaken the sunken assets, offset the operation cost and extend the economic life.

2.2.4. Abundant market of oilfield geothermal utilization

Oilfields not only act as energy producers, but also energy consumers. A large amount of heat, by burning oil, gas and coal, is consumed in oilfield for living heating, thermal recovery of heavy oil, oil gathering and transportation (Wang et al., 2016). For example, heating is required for crude oil dehydration and transportation, and especially heavy oil must be heated to reduce its viscosity before pipeline transportation. However, it involves a considerable energy consumption and operation cost, such as only two crude oil gathering station in north China could cost as high as \$503,000 per year on electricity and gas consumption (Liu et al., 2013). Large demand of thermal energy brings abundant market of oilfield geothermal utilization, especially in aging oilfields. Harvesting oilfield geothermal energy to supply oilfield consumption could compensate the overall operation cost.

2.2.5. Government and company support

The efforts of government and oil companies have made favorable environment and exceptional opportunity to boost the geothermal utilization. For example, China is actively formulating incentive policies and increasing investment to facilitate the development of geothermal energy. According to China's 13th Five-Year Plan, annual geothermal utilization should reach 50 million tons of standard coal equivalent by the end of 2020, and by that time an integrated technological and industrial system for the development and use of geothermal resources would be in place across the country. Oil and gas companies, which generally focused solely on pumping oil and gas for decades, have seen the potential of geothermal resource and started to look for the opportunity to generate geothermal energy from existing wells of their own, such as Continental Resources, Denbury Resources, and Hilcorp Energy, which have done geothermal testing during routine production to recover both hydrocarbon and heat (Erdlac, 2010).

As discussed above, the built-in advantages that oilfield features to develop geothermal energy indicates a promising future of oilfield geothermal resource utilizations to meet the thermal energy demand, extend economic life and offset the operation cost.

3. Methods of extracting oilfield geothermal resource

As a type of energy stored in subsurface formation, oilfield geothermal energy need to be extracted before utilization. In current practice, oilfield geothermal resource is mainly harvested by a liquid medium from existing oil wells and utilized in form of hot fluid. The liquid medium could be produced water from an active production well, or working fluid injected to and circulated out from an abandoned well. These two main methods of geothermal energy extraction are discussed as follows.

3.1. Geothermal extraction by produced water from producing wells

This method takes advantage of geothermally heated formation water, which is a byproduct of oil and gas production. As shown in Fig. 2, high temperature water is produced to surface together with hydrocarbon and then water will be separated and sent to processing facilities to capture the heat, either distributed for direct use, or

converted to electricity by power plant, fulfilling the coproduction of hydrocarbon and geothermal energy.

The producing well used for geothermal extraction and utilization are mainly referred to high water production wells. Massive water production is reported both in mature oilfields (Xin et al., 2012), and in some unconventional plays (Kondash et al., 2017). An average of 25 billion barrels of water are produced annually from oil and gas wells within the United States, and only in Texas 7.4 billion barrels of water are produced per year, which accounts for 30% of the total volume of produced water generated in the United States (Liu et al., 2015b). Historically, produced water has been an inconvenience and required costly disposal; however, it is now being looked at as a resource that can be used to produce electricity for field operations or sold to the grid. Tester et al. (2006) estimated the power generation by coproduced water in Oklahoma and Texas ranges from 2072 to 9965 MW. Mining the geothermal energy from these high temperature water could deliver near-term energy savings, diminish greenhouse gas emissions, and offset the expense of costly water disposal.

However, the geothermal utilization performance of this method is directly tied to the oil and gas production operation, and highly relies on the production rate and water temperature. Such a dependency will result in a corresponding shutdown or decrease in geothermal utilization when the well is shut down or decrease in production rate, especially for geothermal power generation, which has strict requirements on water flow rate and inlet temperature. It is estimated that currently available technologies typically require at least 15,000 barrels of water per day with a minimum temperature of 101.7 °C for efficient and economic power generation (Liu et al., 2015b). Actually, the water rate and temperature threshold value demonstrate the limitations of oil well selection for geothermal power generation, rather than value of thermal recovery.

In addition to production rate and temperature, chemistry of produced water should be taken into consideration before pumping produced water to power plant for power generation. Scaling and corrosion may happen in the heat exchanger system as a consequence of pressure and temperature change, which will increase the operating and maintenance cost and reduce the economic benefits (Jamero et al., 2018; Liu et al., 2015b). Each project of produced water power generation may vary on specific cases, but these problems must be taken into considerations and meanwhile they set major limitations the selection of candidate wells for geothermal production.

3.2. Geothermal extraction by working fluid from abandoned wells

An alternative approach is to recover geothermal resources through abandoned oil wells. After hydrocarbon reservoirs are depleted under economic level, the wells will always be abandoned. These wells are usually regarded as enduring liabilities, which lead to significant costs to plug and abandon. However, existing wellbore is an access to subsurface geothermal resource, therefore an abandoned well could provide an opportunity for geothermal heat extraction. The concept of this method is to select and repurpose the depleted petroleum reservoir to geothermal reservoir and harvest the heat by working fluid injected from surface.

Since there are no geofluids supplied by the pay zones in abandoned wells, working fluid is needed to be injected to the targeted depth, playing the role of heat extractor and heat carrier. The common practice is injecting fluid at the surface, and the fluid will be gradually heated up by the surrounding formation as it flows down. When the injected fluid reaches the bottom of heat exchanger, where it gains the maximum temperature, the fluid changes the direction, flows upward and ascends to the wellhead. Return fluid will be collected at the wellhead and heat will be captured and utilized for different purposes. To recover heat from abandoned wells, some wellbore retrofitting may be needed before injection fluids. U-tube and double pipe heat exchangers are two common types of retrofitting to abandoned wells.

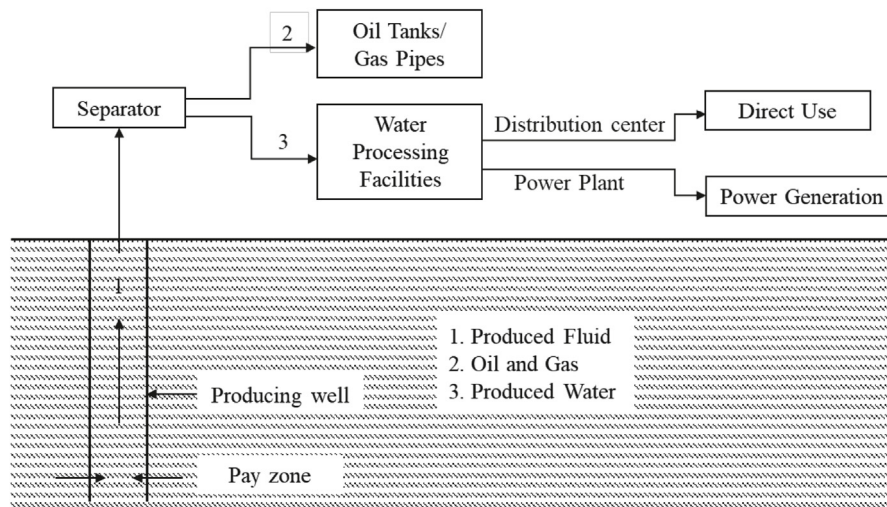


Fig. 2. The Schematics of produced fluid flow for geothermal extraction from producing wells.

3.2.1. U-tube heat exchangers

U-tube heat exchanger is placed in the wellbore prior to filling the well with certain material with a desirable thermal conductivity to improve heat transfer from formation to the heat exchanger. Typically, a working fluid is circulated inside the closed loop of the u-tube heat exchanger extracts the geothermal heat from the reservoir and flows back to the surface (Fig. 3).

3.2.2. Double-pipe heat exchanger

Another method is to construct the wellbore as a coaxial double-pipe heat exchanger, where the casing acts as the outer pipe and the tubing as the inner pipe. The working fluid is injected to the annulus between casing and tubing and flows down towards the targeted depth while extracting heat from surrounding formation. After the fluid reaches the bottom hole, it flows upwards through tubing to the well-head. The tubing is always thermally insulated to reduce the heat

transfer between fluid in tubing and fluid in annulus. Return fluid will be collected at the wellhead and heat will be captured and utilized for different purposes. Downhole constructions and fluid flow path are displayed in following Fig. 4.

In practice, u-tube heat exchanger has been proven as the most common form used to extract heat from shallow subsurface for space heating and cooling. In literature, recent research has started to switch the attention to double pipe heat exchangers due to the advantages of double pipe in heat exchange efficiency, saving pumping energy and less usage of grout. Compared u-tube heat exchangers, double pipe heat exchangers have a larger surface area to exchange heat and containing a higher volume of fluid through to exchange heat. Under same injection rate, fluid flow velocity in double pipe geometry could be lower, which needs less hydraulic pressure to circulate the fluid, resulting in a lower pumping energy consumption. Additionally, in cased abandoned wells, constructing a double pipe heat exchanger is a better choice than

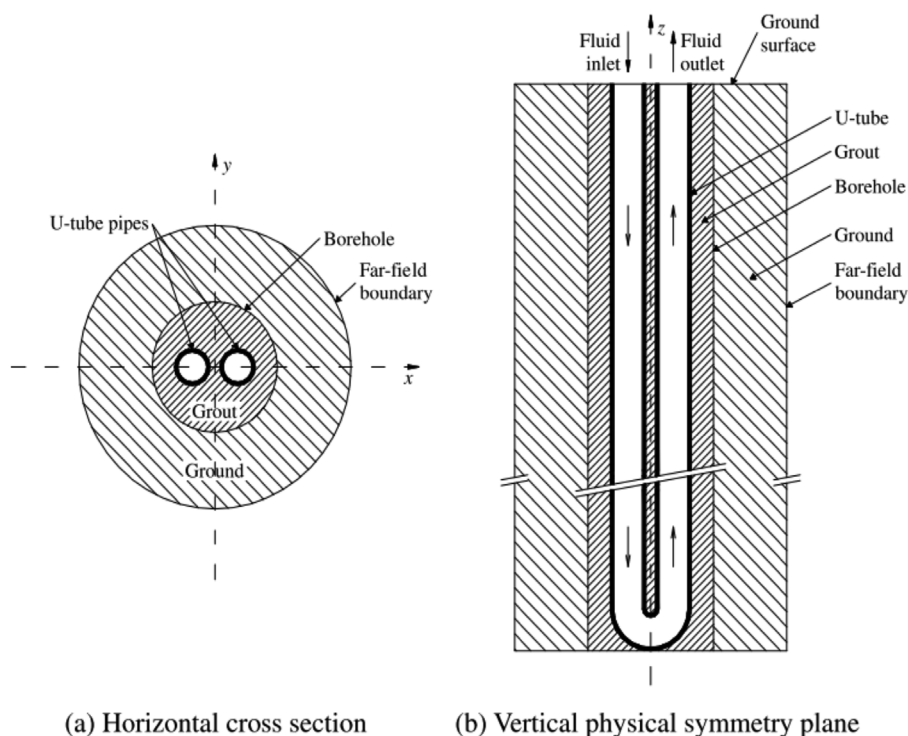


Fig. 3. Schematic of a U-tube heat exchanger in an abandoned well (Li and Zheng, 2009).

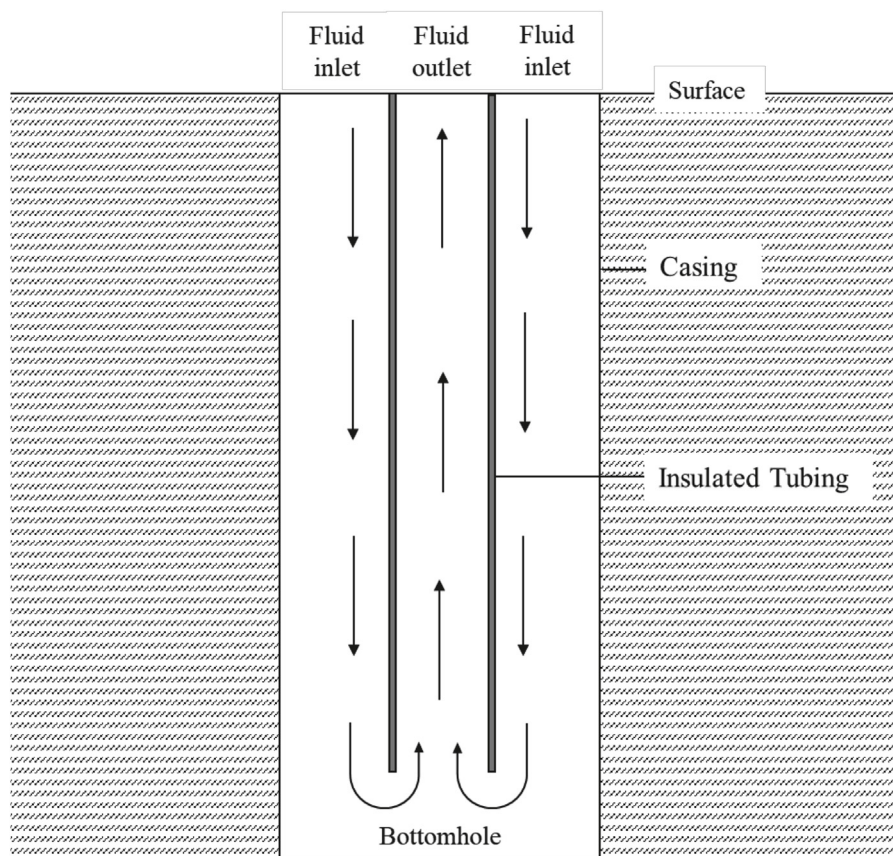


Fig. 4. Schematic of a double pipe heat exchanger in an abandoned well.

u-tube, because the outer pipe (casing) is already present, saving money and time. Finally, coaxial geometry of double pipe heat exchanger has advantages in reducing the thermal resistance between the circulating fluid and the wellbore.

Scholars have conducted lots of research to assess the potential of geothermal production from abandoned wells on both u-tube and double pipe heat exchanger techniques (Bu et al., 2012; Davis and Michaelides, 2009; Garbai and Mehes, 2011; Kujawa et al., 2006; Li and Zheng, 2009; Lyu et al., 2018, 2017; Noorollahi et al., 2015; Templeton et al., 2014; Zheng et al., 2011). These researches established different mathematical models to estimate the heat extraction, conducted case studies and testified the geothermal potential in abandoned wells. Although each case was distinctive, key influencing parameters are identified by the literature as injection fluid temperature, injection rate, insulation type, bottomhole temperature and fluid selection.

3.3. Discussion on geothermal extraction methods

Different from geothermal extraction from producing wells, in which produced water is dominated by oil production, heat extraction in abandoned wells is more manageable with flexibility for adjustments, such as injection fluid selection, injection rate and injection fluid temperature. This flexibility enables the operator to manage and control the whole process of geothermal extraction.

In addition to the difference of manageability, the essential difference of two geothermal extraction methods lies in the source of geothermal supply. Active production well could supply produced water not only at high temperature, but also at sufficient production rate. Therefore, along with the continuous oil production, the produced water could provide long-term lasting geothermal energy input (in terms of volume and temperature) for power generation. On the other hand, the working fluid injected to the abandoned well only get heated

during the circulation in the wellbore, although it could be managed to reach similar temperature as produced water, but the flow rate is far from “competitive” to a producing well. The abovementioned difference will lead to corresponding differences in geothermal utilization, which will be addressed in the following section.

4. Methods of utilizing oilfield geothermal resource

Once the geothermal energy is extracted and brought to surface, the hot fluid will be pumped to processing facilities for different utilizations determined by the level of temperature and flow rate. Direct use (no energy conversion process) and power generation (thermal to electricity conversion) are the main methods of oilfield geothermal utilization, and adapted combined heat and power system is also covered in this discussion.

4.1. Geothermal direct use

Direct use of geothermal resource is arguably the oldest and most versatile utilization since it has been widely applied in more than 82 countries around the world for decades (Lund and Boyd, 2016). Direct use of oilfield geothermal could be divided into two categories. One is the traditional direct-use applications, such as heating buildings, greenhouses planting, drying crops, and several industrial processes. The other is special oilfield applications, including oil gathering heat tracing, crude oil transportation and geothermal water flooding.

To reduce the viscosity of crude oil, hot fluid circulation is usually used for crude oil gathering and transportation by burning oil and gas. Hot fluid could be pumped into the heat-trace pipe attached to the pipeline, with insulation installed around the whole. In mature oilfields, where massive hot water is produced, produced water could be used for crude oil gathering heat tracing. Replacing the oil/gas-burning heated

water by geothermally heated water could save considerable fresh water and reduce the crude oil transpiration cost.

Geothermal water flooding is an important oilfield geothermal direct application, especially for heavy oil enhanced recovery. It has been proved that hot water flooding can decrease oil viscosity and mobility ration, and then improve ultimate oil recovery (Goodyear et al., 1996). Current practice is to heat water by burning oil or using electricity heating, which leads to high operation cost. Comparing to the conventional hot water flooding, geothermal water flooding can carry the heat of high temperature fluid from deeper aquifer into shallower oil formations effectively with the method of “deep production and shallow injection” in the same well (Liu et al., 2015a, b). Geothermal waterflooding has been experimentally and mathematically studied and showed a 4%–10% recovery improvement over conventional water flooding (Pederson and Sitorus, 2001; Wang, 2012). Furthermore, when conducting geothermal water flooding project, injection rate and temperature needs to be properly designed in order to avoid formation damage, which will be elaborated in the later section. In general, geothermal water flooding is not only a method to utilize geothermal energy, but also definitely helpful to enhanced oil recovery.

4.2. Geothermal power generation

Since the oilfield geothermal resource is classified as intermedium to low-temperature geothermal energy, binary cycle generation of Organic Rankine Cycle (ORC) is usually used for oilfield geothermal power generation (Liu et al., 2015a, b; Tester et al., 2006; Wang et al., 2016). Binary power generation is a well-established technology for utilizing low to moderate temperature geothermal fluids (DiPippo, 2015). As shown in Fig. 5, electricity is generated through a binary which transfers heat from the hot fluid to a secondary fluid that vaporizes at a lower temperature and higher pressure, and the vapor is then used to drive a turbine to produce electricity (Liu et al., 2015b). Boyle (2012) pointed out that the binary power plants could minimize the gas emission due to the closed loop operation and extend the range of potential geothermal resources.

4.3. Adapted combined heat and power (CHP) systems

In geothermal power plant, only 10% of the energy from the produced geothermal fluid can be converted to electricity (Zarrouk and Moon, 2014), resulting in residual thermal energy in the discharged water, sometimes considered as waste heat. Combined heat and power

(CHP) systems is one type of distribution energy system, which could offer an effective way to make use of the waste heat energy from a power plant for building heating, combining electricity generation and heat delivery to a nearby district heating system (Zhang et al., 2016).

Specifically, for oilfield application, an adapted CHP system could provide an integrity solution of geothermal power generation and direct utilization. Li et al. (2014) proposed an adapted CHP system for oilfield application, which included a binary power plant for power generation, series of heat recovery units to process the discharged water for direct use (Fig. 6). A portion of the generated electricity could be used to meet the demand of the system consumption, and the remaining merges into the grid. The heat recovered from discharged water could be used for nearby space heating and crude oil gathering and transportation.

Li et al. (2014) demonstrated the promising potential of this adapted CHP system by case study in oilfield in China. The system outputted a net power of 569 kW and could serve twenty 5000-square-meter residential buildings for space heating and more than 8000 people for domestic hot water supply. Economically, the payback time for the total geothermal system is only about 0.8years, which is very attractive in engineering applications.

To sum up, oilfield geothermal utilizations are typically identified by associated resource temperature and flow rate. Power generation always requires relatively higher temperature as well as higher flow rate than direct utilizations, in order to meet the requirement of energy conversion. However, there is no distinct temperate and flow rate to distinguish these two types of utilizations. Direct utilizations could also take place in higher temperature range under high flow rate. To improve the energy efficiency and maximize the thermal recovery, geothermal power generation and direct use can be combined by adapted CHP system.

5. Current development of oilfield geothermal utilization

After reviewing the abundant reserve, unique advantages, methods of extracting and utilization oilfield geothermal resource, noteworthy progresses of worldwide oilfield geothermal utilizations are highlighted.

5.1. Oilfield geothermal direct utilization

Among the entire worldwide geothermal direct utilizations, geothermal energy extracted from oilfield are reported from a number of countries from different parts of the world. As Lund and Boyd (2016)

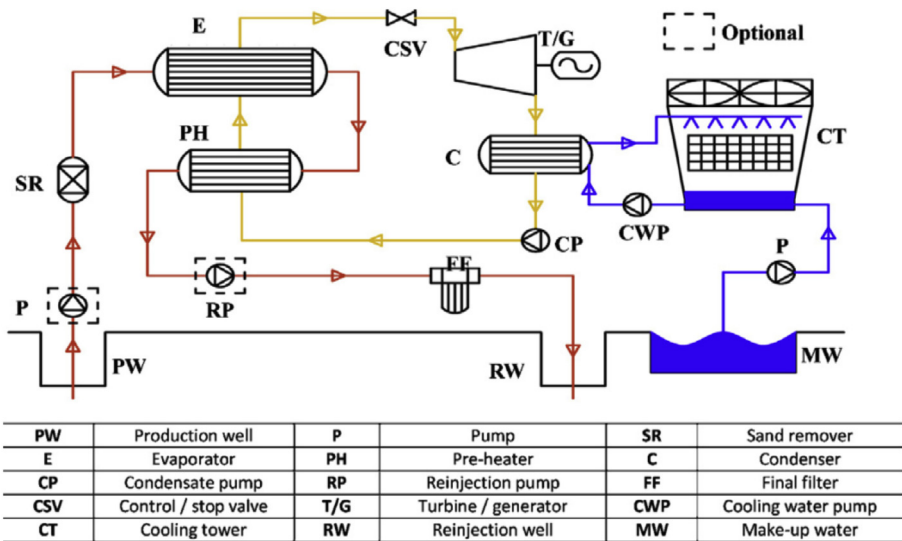


Fig. 5. The schematic of a basic binary geothermal power plant (Zarrouk and Moon, 2014).

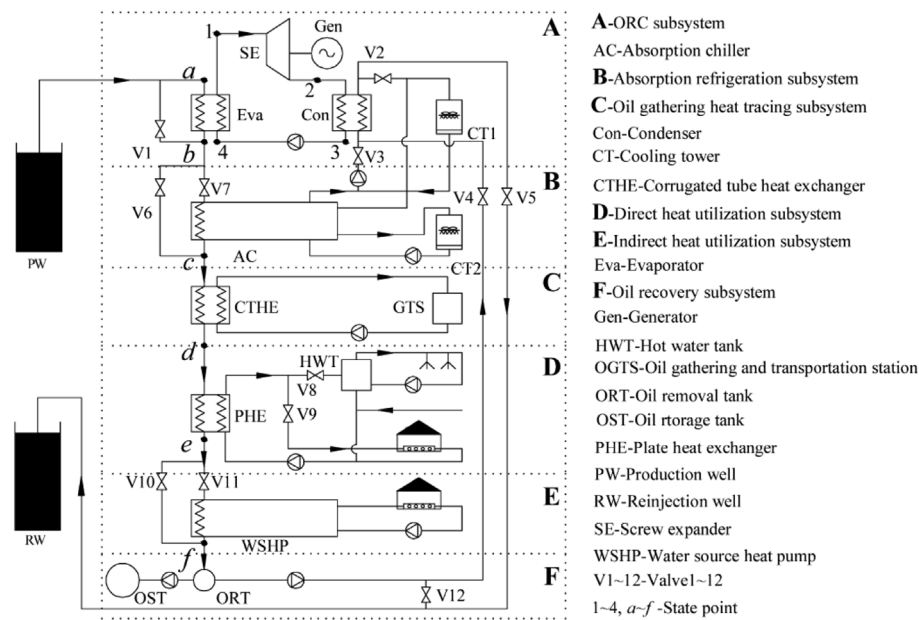


Fig. 6. Schematic diagram of the adapted CHP system (Li et al., 2014).

reported, since 1970s, Austria has started to use abandoned oil exploration wells to produce thermal waters, which were used at the spa resorts. Similarly, Albania also utilize its low temperature geothermal resource (up to 65 °C) from abandoned wells for greenhouse heating. In Hungary, geothermal water is used for thermal water flooding for the secondary oil production technology in the Algyő oilfield. Another application of geothermal water is that gathering pipes are heated by thermal water in the heavy oil producing oilfield Sávoly in the Southwest of Hungary.

China is promoting the use of geothermal power in heavily air-polluted north region, especially in the “Beijing-Tianjin-Hebei” region, to replace coal for heating to reduce carbon dioxide emissions and improve air quality. Several oilfield geothermal direct-use projects have been conducted in that area, covering a wide variety of direct-use applications.

Furong residential area in Shengli Oilfield, which locates in the northeast China, has been benefited a nearby house heating station supplied by oilfield geothermal water since 2002. In total, geothermal heating in this residential area has already saved up to 10.3 EJ energy, which is equivalent to 3×10^4 t of standard coal and 2×10^4 ton of oil, and reduced 9.8×10^4 ton of CO₂ emission. According to Wang et al. (2016), geothermal space heating using moderate formation depths (200–3000 m) reached 40×10^6 m², accounting for 40% of the total geothermal heating area in China, which replaced 1.2×10^6 ton of standard coal per year and reduced 3×10^6 ton of CO₂ emission per year.

Oilfield geothermal water is also used for heat-trace oil gathering and crude oil transportation in China. Two abandoned wells in Huabei Oilfield have been reconstructed to produce geothermal water at 600 m³/day and 100–110 °C for heat-tracing gathering and transportation, which can save about 5 ton of oil and 3500 m³ of gas every day (Wang et al., 2016).

Table 2 summarizes worldwide oilfield geothermal direct utilization projects and we could tell from the table that oilfield geothermal direct utilization projects were conducted in both producing wells and abandoned wells. General direct utilizations of oilfield geothermal resource have been successfully and widely applied by the courtesy of mature geothermal direct use technology and practices. The specific oilfield geothermal direct use is mainly conducted in the oilfields which produce high viscous oil.

5.2. Oilfield geothermal power generation

United States is one of the leading countries on research and utilization of low-temperature geothermal power generation. McKenna et al. (2005) assessed that over 1000 MW electric power could be created from produced water along the Gulf Coast. A gas well in Texas was reported to be capable to produce 340 kW net power using coproduced water (Sanyal and Butler, 2010). Bennett et al. (2012) states the net power output from geothermal power generation in mature oilfields in Los Angeles basin would be around 7430 kW. These preliminary assessments discovered the great potential and laid a solid foundation for oilfield geothermal power generation. Soon afterwards, oilfield geothermal power generation projects conducted in the fields.

Department of Energy (DOE) carried out a power generation pilot project with low-temperature geothermal power associated with produced water beginning in 2006 (Nordquist and Johnson, 2012). The Organic Rankine Cycle power station used 90.6–98.9 °C produced water at production of 40,000 barrels/day, and generated 180 kW electricity (net power generation of 132 kW). Gosnold (2017) reported another successful case of coproduction project in North Dakota. Researchers at the University of North Dakota successfully generated geothermal power from 98 °C water which was pumped from waterflood wells in the Williston Sedimentary Basin in western North Dakota. This project is reported as first commercial geothermal power production from an oil and gas well in USA.

Besides DOE, efforts also have been made by oil and gas companies. Erdlac (2010) mentioned that five companies recognized the geothermal resource in oil and gas wells and made efforts to harness this energy for electrical power generation in operations use. The operations were taken place in North Dakota, Summerland Field in Mississippi, south Louisiana, Texas Raton Basin of Colorado, attempting to use wells with production rate at 4000 to 5000 barrels per well per day range for power generation in order to offset operating expense of oilfield.

Another well-known project of oilfield geothermal power generation was reported by Xin et al. (2012), as the first geothermal power plant built in China using produced water from waterflooding wells. A 400-kW power generator was installed and started to generate electricity using 110 °C hot fluids at 2880 m³/d coproduced from Liubei oil reservoir in Huabei Oilfield. By the end of 2011, the cumulative energy generated was about 31×10^4 kWh. Table 3 summarizes the implemented projects of oilfield geothermal power generation in a binary

Table 2
Summary of worldwide oilfield geothermal direct utilization projects.

Oilfield Location	Geothermal Resource	Type of Utilizations	Remarks	Reference
Albania	Abandoned Wells	Greenhouse Heating	Geothermal water (up to 65.5 °C) was used for greenhouse heating	Lund and Boyd, 2016
Styria, Austria	Abandoned Wells	Spa Resorts	Geothermal waters that were used at the spa resorts Loipersdorf and Waltersdorf	
Algyő, Hungary	Produced water	Geothermal Water EOR	Geothermal water has been used for thermal waterflooding since 1969	Liu et al., 2015a, b
Sávoly, Hungary	Produced water	Heat-trace Oil Gathering	Geothermal water was used to heat the gathering pipe in Sávoly	
Shengli, China	Abandoned wells	Spacing Heating	Space heating in Furong area saved 3×10^4 t of standard coal and 2×10^4 ton of oil from 2002 to 2015	
Daqing, China	Produced water	Spacing Heating	5 projects replaced 7000-ton standard coal per year	Wang et al., 2016
		Crude Oil Transportation		
Liaohe, China	Produced Water	Spacing Heating	12 projects replaced 24,400-ton standard coal per year	
Huabei, China	Abandoned Wells	Heat-tracing Oil Gathering	Geothermal oil gathering from 2 abandoned well save about 5 ton of oil and 3500 m ³ of gas every day	
Zhongyuan, China	Produced Water	Spacing Heating	One project saved 2537 ton of standard coal per year	

power plant using produced water.

It is worthy to mention that current oilfield geothermal power generation projects are all conducted using produced water. Although it is more manageable to extract geothermal resource from abandoned wells, operators all choose producing wells to initiate the project. As discussed in section 3.3, produced water could provide hot water at sufficient rate and temperature for power generation, while geothermal recovered from abandoned wells are currently reported for direct use.

6. Challenges in oilfield geothermal resource development

Although oilfield geothermal resource is gaining increasing attention and have achieved notable development, there still exist challenges restricting quick and further development of the oilfield geothermal resource. The key factors limiting oilfield geothermal energy exploitation and utilization are identified as low energy conversion efficiency, insufficient involvement, inadequate assessment and planning, laws and regulation, and other problems.

6.1. Low conversion efficiency

Technically, the biggest factor that hampers the geothermal energy utilization is the low efficiency of power generation, especially for low to medium temperature geothermal resource. The low conversion efficiency will make geothermal power generation less competitive than other forms of power generation power plants, such as coal, natural gas, oil, and nuclear power stations (Zarrouk and Moon, 2014).

Moreover, as one of important method of oilfield geothermal production, power generation using coproduced water may bring potential hazards to further deteriorate the performance of power generation and the economic competitiveness. First, the dynamics of coproduced water rate and temperature could vary as the field is developed. If coproduced water fails to input sufficient heat, power generation performance will be affected. Secondly, coproduced water may cause scaling and corrosion problems for heat exchangers, which will increase the maintenance cost and reduce the heat exchanger efficiency, resulting in the decrease of power output and economic competitiveness.

Table 3
Notable implemented oilfield geothermal power generation projects.

Location	Flow Rate, bbl./d	Water Temp., °C	Net Power Generation, kW	Remarks	Reference
Wyoming, USA	40,000	90.6–98.9	132	Produced over 2,210 MWh of power in the first 3.5 years	Nordquist and Johnson, 2012
North Dakota, USA	30,000	98	250	First commercial project of geothermal power generation in an oil and gas well in USA	Gosnold, 2017
Huabei, China	18,114	110	310	Generated was about 31×10^4 kWh by the end of 2011	Xin et al., 2012

6.2. Insufficient involvement, assessment and planning

Compared to oil and gas exploration and production, and traditional geothermal field development, there are still relatively fewer activities engaged in oilfield geothermal exploitation and utilization, which slows down the growth of oilfield geothermal utilization. More efforts should be put into scientific and technological research on deepening oilfield geothermal resource exploration, assessment and development. In return, progress in oilfield geothermal development made by the researchers will attract and bring more involvement, which could create a virtuous circle and further boom oilfield geothermal industry.

Hydrocarbon reservoir geothermal potential is well assessed in some areas and oilfields, such as Texas (Erdlac et al., 2007), Michigan and Illinois Basins in USA (Crowell and Gosnold, 2014), Daqing Oilfield, and Bohai Bay Basin in China (Wang et al., 2016). However, there still exist lots of areas and oilfields across the world, whose geothermal potential haven't been systematic and comprehensive assessed. Meanwhile, the current oilfield geothermal operators are still in the attempt of developing geothermal resource from their own wells. A long-term blueprint and rational flowchart to develop large-scale geothermal resource are still blur. Accurate assessment and planning will help geothermal producers make best decision on each step of geothermal development and achieve best economic and social benefit. To the contrary, unclear understanding of geothermal resources directly limits the exploration, exploitation, and utilization of geothermal resources.

6.3. Laws and regulations

According to Airhart (2011), the estimated cost of oilfield geothermal development is cheaper than solar and nuclear, and comparable to wind, but more expensive than coal and natural gas. Essentially, oilfield geothermal industry is a new industry with relatively low profits. Although governments have made incentive policy, to further boost and ultimately promote the vigorous development, industrial policies, finance and tax policies, and other aspects should be more detailed, specialized, favorable and complete.

To be specific, laws and regulations should be created specifically designed to accommodate the uniqueness of oilfield geothermal

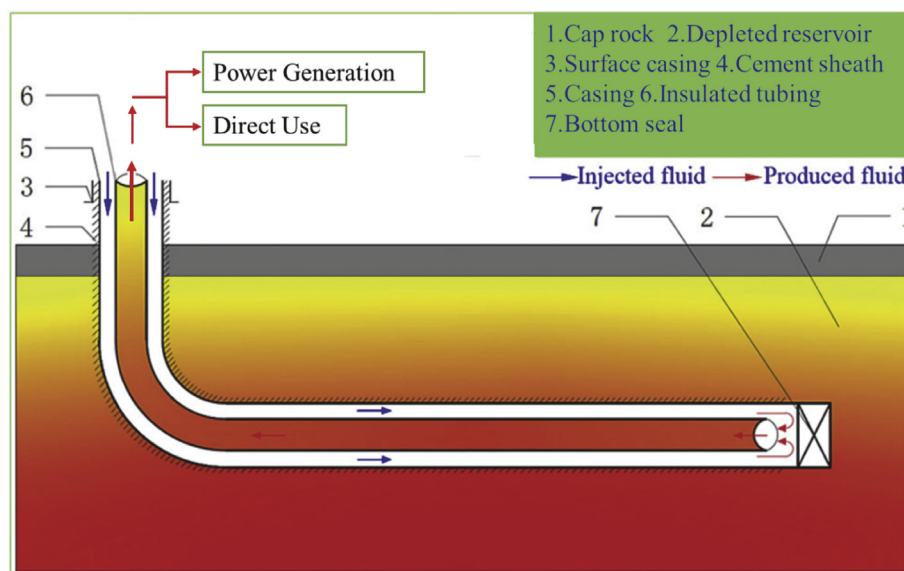


Fig. 7. The schematics of oilfield geothermal utilizations from a horizontal well (After Cui et al., 2017).

resources. Economically, regulations should provide best protection to ownership interests. From environmental and social point of view, regulations should be made to keep geothermal resource extracted in sustainable manner. A law designed with these factors will not only encourage geothermal heat utilization and production, but will also make sure these resources remain viable and renewable for a long time.

6.4. Formation damage induced during geothermal energy extraction

The decrease of reservoir temperature not only interrupts the in-situ thermal-chemical equilibrium, but also introduces mechanical formation damage processes in subsurface (Yuan and Wood, 2018). To develop oilfield geothermal resource, keeping enough high rates of fluids circulation is essential to guarantee the exploitation of geothermal systems economically viable. However, the conditions of high flowing rates and versatile temperature distributions can trigger the significant decline of well injectivity and productivity by exaggerating the transport and plugging of clay particles and permeability impairment (You et al., 2015). The influencing factors on the migration and plugging of clay particles mainly consist of physic-chemical properties of clay particles, fluid pH and salinity, temperature, water film (adsorption water and osmosis water) and fluid flowing rates (Tchistiakov, 2000). Therefore, the assessment and quantification of fines-migration-induced formation damage is a vital part of the feasibility study for oil-field geothermal projects.

In addition, the temperature-dependent rock properties significantly affect the heat recovery in oilfield geothermal development, including changes of rock storability, electrical conductivity, seismic properties, and properties of in-situ fractures in reservoirs (Jaya et al., 2010; Kristinsdóttir et al., 2010; Milsch et al., 2010). The permeability impairment on inflow performance of oils/gas wells can bring significant negative effects in geothermal extraction (Ghassemi et al., 2005).

6.5. Other problems

Locations of geothermal production well and power plant may have impact on direct use and power generation. Except those wells that geothermal energy is utilized nearby in forms of heating oil tanks or power wellsite equipment, long distances from wells to power plant and high demand areas will limit the direct use of geothermal fluid and power generation. Most wells are located in low residential areas, where the demand in heating and electricity is low. Therefore, long

distance transportation could increase the operation cost, resulting in geothermal utilization less cost-effective. Thus, special consideration should be given in candidate well selection.

Another potential challenge lies in the abandoned wells. Some wells may have been abandoned for years with high potential in malfunctions after re-engineered for geothermal production, possibly in forms of failure in double pipe installation, loss of casing integrity and wellhead malfunction, which may lead to increase in nonproduction time and suspend in geothermal utilization. Therefore, proper inspections should be made for selected wells before retrofitting for geothermal production, and keep routine maintenance after production.

7. Advanced researches and emerging techniques

Topics related to oilfield geothermal utilization have been researched for decades. With the goals to unlock the oilfield geothermal potential, enrich the heat recovery methods, maximize recovered thermal energy and obtain power generation in sustainable manner, scholars have done prolific and effective research work, covering a wide range of topics from oilfield geothermal resource assessment at exploration stage to last step of power generation performance. Meanwhile, some emerging techniques are considered to be the possible solution to speed up the development of oilfield geothermal utilization. Selections of remarkable research and emerging techniques are outlined in this section.

7.1. Geothermal extraction from abandoned wells

It is notable that all the previous discussion about downhole heat exchangers are conducted in vertical abandoned wells. Zhang et al. (2014), Feng et al. (2015) and Cui et al. (2017) are the few preliminary works on using horizontal wells for geothermal exploitation (Fig. 7). Horizontal well brings significant boom for oil and gas productions by enlarging reservoir contact area, and ultimately increase the production of oil and gas (Yuan et al., 2017). By the same principle, geothermal production could be benefited using abandoned horizontal well as an enhanced thermal recovery method. Under a same flow rate, the injected fluids have shorter contact time with the highest temperature formation at the bottom-hole depth, while in a horizontal well, injected fluid could be continuously heated up when flowing through a long horizontal lateral of highest temperature, which makes the return fluid contain larger amount of thermal energy. The heat extraction efficiency

per unit mass flow will be improved.

Zhang et al. (2014) studied the feasibility of generation electricity from hydraulic fractured horizontal well by circulating produced water, with a case study in Daqing Oilfield in China. Feng et al. (2015) proposed two coaxial heat exchanger configurations to exploit forced convection in a horizontal wellbore and highlighted the improved heat extraction and heat exchange efficiency. Cui et al. (2017) also presented similar method to install double pipe heat exchanger and circulate fluid in horizontal wells for geothermal exploitation and established simulation model to calculate the heat extraction rate. Sensitivity studies were conducted to distinguish the impacts of injection rate, horizontal length, and tubing insulation on extracted amount of thermal energy.

As verified by these works, geothermal extraction using a horizontal well could enhance heat exchange between the formation rocks and working fluids through long horizontal segment, therefore, geothermal extraction from a horizontal well could be considered as an enhanced thermal recovery option to promote the oilfield geothermal production.

7.2. Thermoelectric generation technology

One of the emerging technologies that is considered to speed up geothermal power growth is the technology of thermoelectric generation (Li et al., 2015). The thermoelectric power generation is based on the Seebeck effect. Under certain temperature difference, thermoelectric modules could directly transform thermal energy into electricity without through mechanical work to turn the turbines (Fig. 8). The application of thermoelectric technology has been recently investigated by a couple of oil and gas industry researchers. The feasibility of oilfield power generation from surface (Liu et al., 2014) and subsurface (Wang et al., 2017) are both illustrated, and the key factors on thermoelectric generation performance are identified (Li et al., 2017).

Liu et al. (2014) experimentally studied and tested the thermoelectric generation technology in geothermal application. Li et al. (2017) modeled the effect of module size and material property on thermoelectric power generation, providing a theoretical basis for the configuration and design optimization of thermoelectric generator and making more efficient use of geothermal resources. Wang et al. (2017) integrated thermoelectric technology with oil production operation and proposed the first design of downhole geothermal power generation in a producing well. By installing thermoelectric devices on the tubing and creating temperature difference between tubing and annulus, electricity is generated from subsurface the thermoelectric device, which eliminates the need of long distance transportation of produced water to surface power plant.

Thermoelectric technology enriches the methods of oilfield geothermal extraction. More importantly, such a technology features a wide temperature range for power generation, as low as 30 °C (Li et al., 2015), which facilitates more geothermal resource and more wells for power generation. Some wells producing insufficiently hot water for binary power generation could be considered as candidate wells for

power generation using thermoelectric technology.

7.3. Nanotechnology

Nanotechnology is science, engineering, and technology conducted at the nanoscale, which is about 1–100 nm. Nanotechnology has a wide range of applications in multidisciplinary, such as molecular devices (Cui et al., 2014; Fang et al., 2011; Ji et al., 2011; Xu et al., 2011), electrical engineering (Ji et al., 2011, 2012; Zhao et al., 2015) and petroleum engineering (Liu et al., 2018; You et al., 2018; Yuan et al., 2017, 2016; 2018a, b). In this paper, nanotechnology is identified to be helpful to accelerate the oilfield geothermal utilization by applying nanostructured thermoelectric material and nanofluids with added nanoparticles.

As abovementioned application of thermoelectric generation technology in oilfield geothermal utilization, the thermometric performance is largely determined by nature properties of thermoelectric material (figure of merit), which is related to Seebeck coefficient, electrical resistivity, and thermal conductivity of the material. To achieve an efficient thermal to electricity conversion, it is favorable to use optimal material with higher figure of merit. And nanoscale and nanostructured thermoelectric materials are promising for increasing figure of merit (Szczeczek et al., 2011).

Wang and Wu (2012) described the efficiency enhancement by using the high performance of nanostructured Bi_2Te_3 -based thermoelectric generator. Szczeczek et al. (2011) also pointed out the enhanced thermoelectric performance of a layered $\text{Bi}_2\text{Te}_3/\text{Sb}_2\text{Te}_3$ nanocomposite at 167 °C. These investigations indicate the promising future of thermoelectric generation technology in oilfield geothermal application, because Bi_2Te_3 -based thermoelectric material is the best selection of materials in the intermediate to low temperature range (Cheng et al., 2016). Therefore as the material science and engineering advances, nanomaterial with higher figure of merit will enhance the power generation performance and further accelerate the application of thermoelectric power generation using oilfield geothermal resource.

Another application of nanotechnology is the utilization of nanofluid for heat extraction. Considering the important role that heat transfer plays in geothermal energy extraction, enhancing heat transfer efficiency is desirable for better oilfield geothermal recovery. One of the major parameters to heat transfer is the thermal conductivity of the working fluid. Improvements in nanotechnology made it possible to add nanoscale particles in a base fluid to enhance the heat transfer (Özering et al., 2010). The fluid where nanomaterials are dispersed into a base fluid is called nanofluid. Addition of high thermal conductivity nanoparticles into the base fluid increases the thermal conductivity of nanofluids, thus significantly enhanced heat transfer capacity compared to traditional working fluids. Sui et al. (2017) investigated the potential of applying Al_2O_3 nanofluid as working fluids to extract geothermal energy from abandoned wells and it turned out that nanofluid is capable to harvest more energy and improve the exploitation of the oilfield geothermal resources, by increasing the returning fluid temperature.

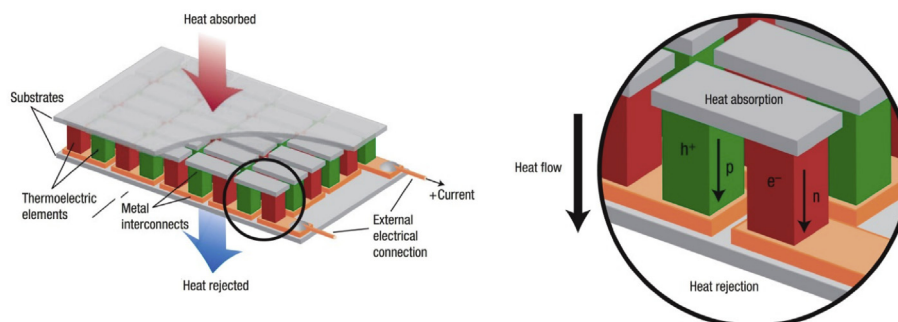


Fig. 8. Schematic of thermoelectric generator (after Snyder and Toberer, 2008).

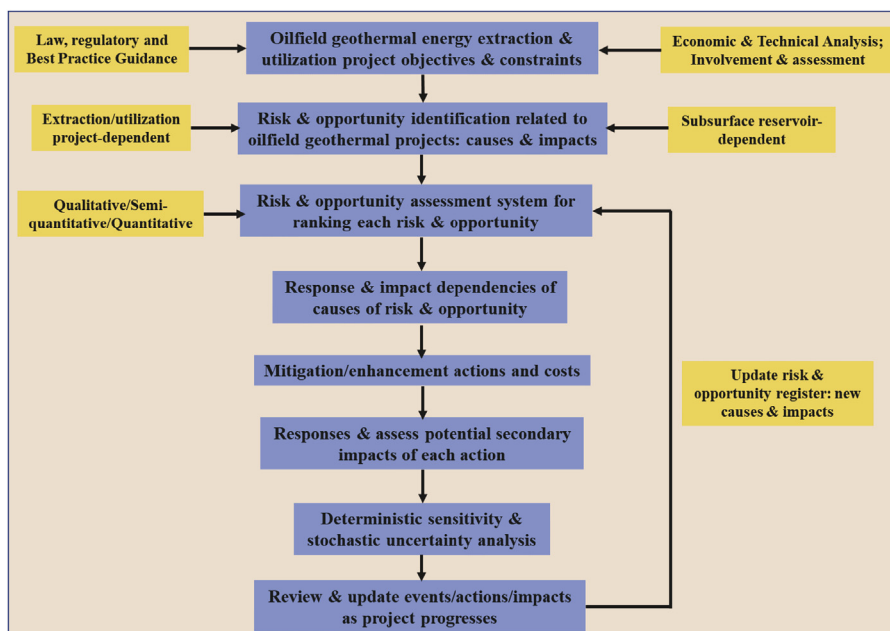


Fig. 9. An integrated workflow for risk and opportunity identification, assessment and management of oilfield geothermal energy projects (after Yuan and Wood, 2018).

8. Integrated risk and opportunity assessment and management

For the geothermal extraction and utilization projects, the integrated risk/opportunity assessment and management requires complex steps conducted in a systematical framework, mainly including identification, assessment and ranking of events and associated actions (Yuan and Wood, 2018). The detailed workflow (Fig. 9) involves integrating qualitative, semi-quantitative and quantitative risk and opportunity assessment of oilfield geothermal energy production actions and other potential risk sources, using a range of analysis tools, and applying performance indicators monitored for rigorous and holistic risk and opportunity management of oilfield geothermal projects.

During oilfield geothermal projects, to identify and recognize risk and opportunity factors, extensive data from laboratory and field data in multiscale (including single well tests, pilot studies involving water injection and production well practice, and associated thermal energy utilization testing) are required (Wood et al., 2008). The types of data recorded for each action and process are likely to be drawn from various laboratory analysis and field measurements, including: core data (rock porosity, permeability, thermal capacity and conductivity, and mechanical properties), well data (phase saturation, reservoir permeability, porosity, heat storability and transfer capacity, and mechanical failures), well injection/production strategies (injection/production heat flux and bottom-hole pressure and temperature) and thermal energy utilization testing (power-generation, or direct-use). The aims of such measurement and analysis is to provide indicators to quantify the scale and extent of risks and opportunities of each actions and processes related to geothermal extraction and utilization in oilfields.

Following the identifications of the scale and extent of various risks and opportunities, it becomes necessary to assess the integrated risks and opportunities of multiple factors performing simultaneously, and identify the top risks and opportunities. Generally, three different approaches can be applied for risk and opportunity assessment (Berner and Flager, 2016), qualitative, semi-quantitative, and quantitative approaches. Through applying the qualitative approach in the two-dimensional matrix of impacts versus likelihood, the operators can prioritize which kind of risk and opportunities require mitigation or enhancement, and then first focus upon the upside-opportunities that are “crucial to capture”. Building upon the qualitative two-dimensional

matrix, the improved semi-quantitative approach can be achieved by introducing a scoring system for two-dimensional matrix and three-dimensional matrix (i.e., impact, likelihood, and duration of specific risks and opportunities), which can provide us more rigor and less subjective assessment. However, both qualitative and semi-quantitative methods cannot quantify consequences of types of actions and processes in terms of both risks and opportunities. For instance, in both qualitative and semi-quantitative methods, the same score in the matrix defined by the events/actions may have different geothermal energy extraction/utilization performance, financial and other outcomes. Therefore, a more quantitative/stochastic analysis should be adopted, where the likelihood of risks and opportunities associated with each event/action is defined as a probabilistic function, and the impacts is selected as actual measurable outcomes (petroleum/heat production rates, and financial indicators and etc.). The stochastic analysis could help determine the confidence levels of outcomes by inputting the diverse likelihoods (probabilities) of specific risks & opportunities of each action applied for oilfield geothermal energy projects. In sum, such rigorous and holistic risk and opportunity assessment workflow is presented in Fig. 9, which aims to improve the decision-making of geothermal extraction and utilization projects.

9. Conclusions

This paper presented a comprehensive review on geothermal resource in oilfield, unique characteristics in the oilfield utilization, current developments, challenges and possible solutions. Based on above review and analysis, the following conclusions can be drawn:

- Utilization oilfield geothermal resource is a beneficial choice to offset the oilfield operation cost, reduce the CO₂ emission and extend the economic life of well and surface infrastructure, especially in the time of low oil prices, aging oilfields, incremental exploration, operating, and decommissioning cost.
- Geothermal development in oilfields not only eliminates the largest barrier of high initial capital costs in drilling, but also offers superior conditions to facilitate geothermal development and benefit the petroleum industry in return.
- Notable progresses have been achieved in a number of countries in

both direct use and power generation, and brought considerable benefit to oilfield and society.

- Despite of certain barriers limiting the development of oilfield geothermal resource, it is expected to see the significant growth of geothermal direct use and power generation after the breakthrough of novel technology and research in the future.

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